EEE104 – Digital Electronics (I)

Lecture 10

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Emerging Electronics and Energy Technologies Research Centre (EEET)

Office: SC242

Office hour: 1:00-5:00 pm, Wednesday

In This Session

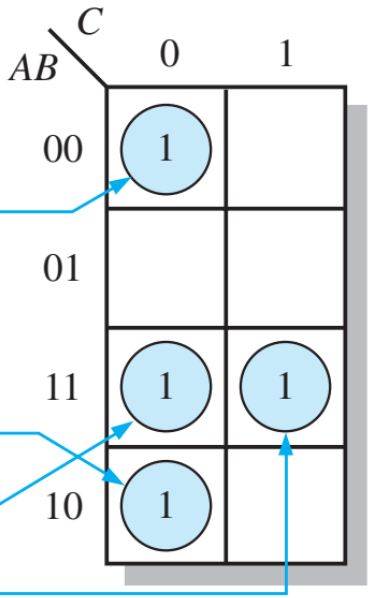
- The Karnaugh Map
- Karnaugh Map SOP Minimization
- Karnaugh Map POS Minimization

Karnaugh Map

- A graphical tool to simplify Boolean expressions.
- It is like a truth table in array form, in which each cell corresponds to a row in the truth table.
- Minimizing Boolean functions using Karnaugh maps is practical only for up to 4-5 variables.

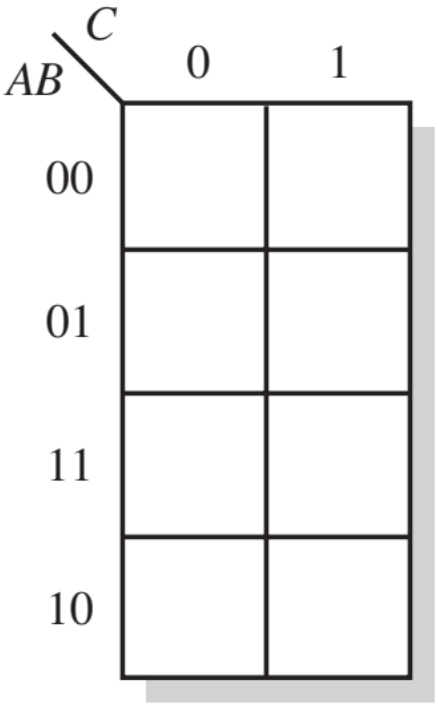
$$X = \bar{A}\bar{B}\bar{C} + A\bar{B}\bar{C} + AB\bar{C} + ABC$$

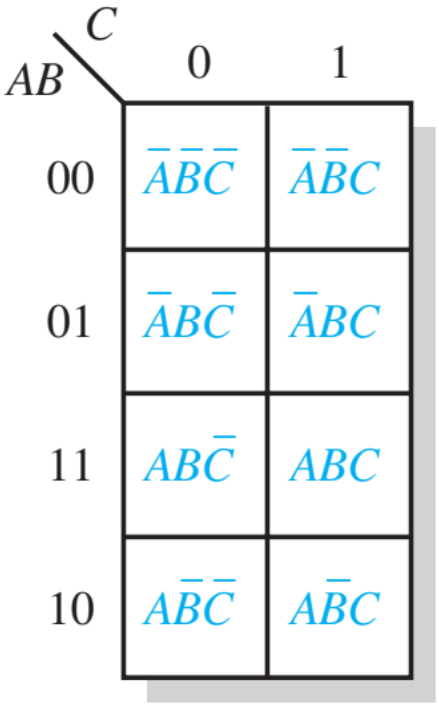
Inputs			Output
A	*B*	*C*	*X*
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1



Karnaugh Map

- The number of cells is equal to 2^n, where n is the number of variables.
- The cells are not arranged according to the magnitude of binary values, e.g. **00 → 01 → 11 → 10 → 00** (Gray-code order)





Karnaugh Map

Cell Adjacency

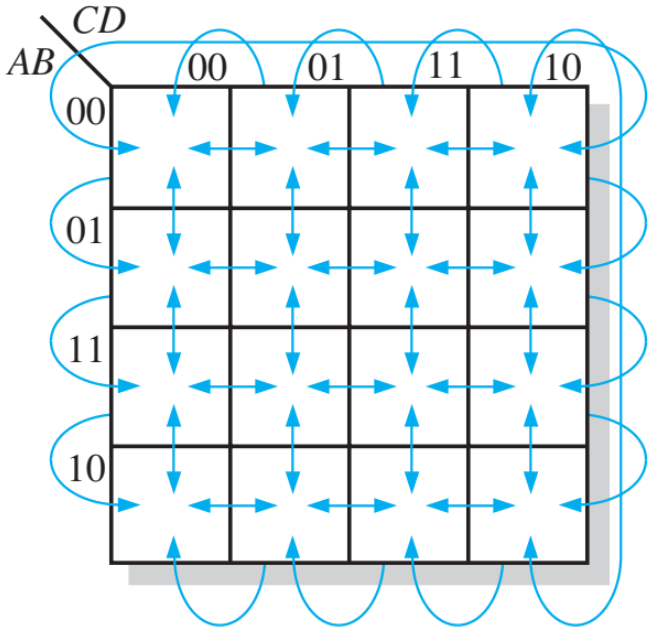
• The cells are arranged so that there is only a **single-variable change** between adjacent cells.

• The binary values of two variables:
00 → 01 → 11 → 10.

• Each cell is adjacent to the cells on its four sides.

• The top row is adjacent to the bottom row.

• The leftmost column is adjacent to the rightmost column. (“wrap-around”)



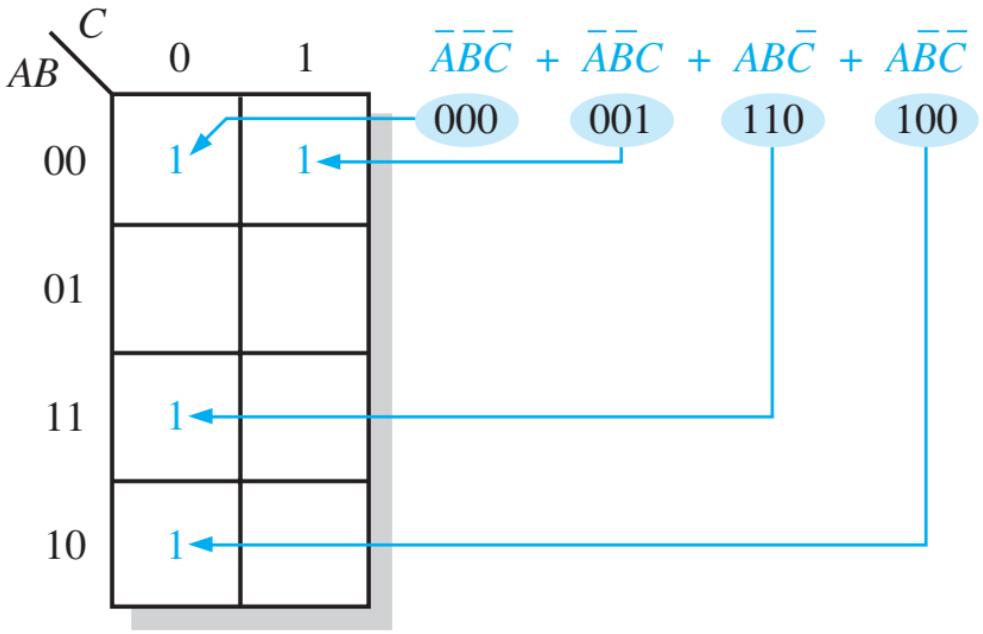
Practice 1

Which pairs are adjacent?
0000/0010, 0000/1000, 0000/0011

Karnaugh Map SOP Minimization

Mapping a Standard SOP Expression

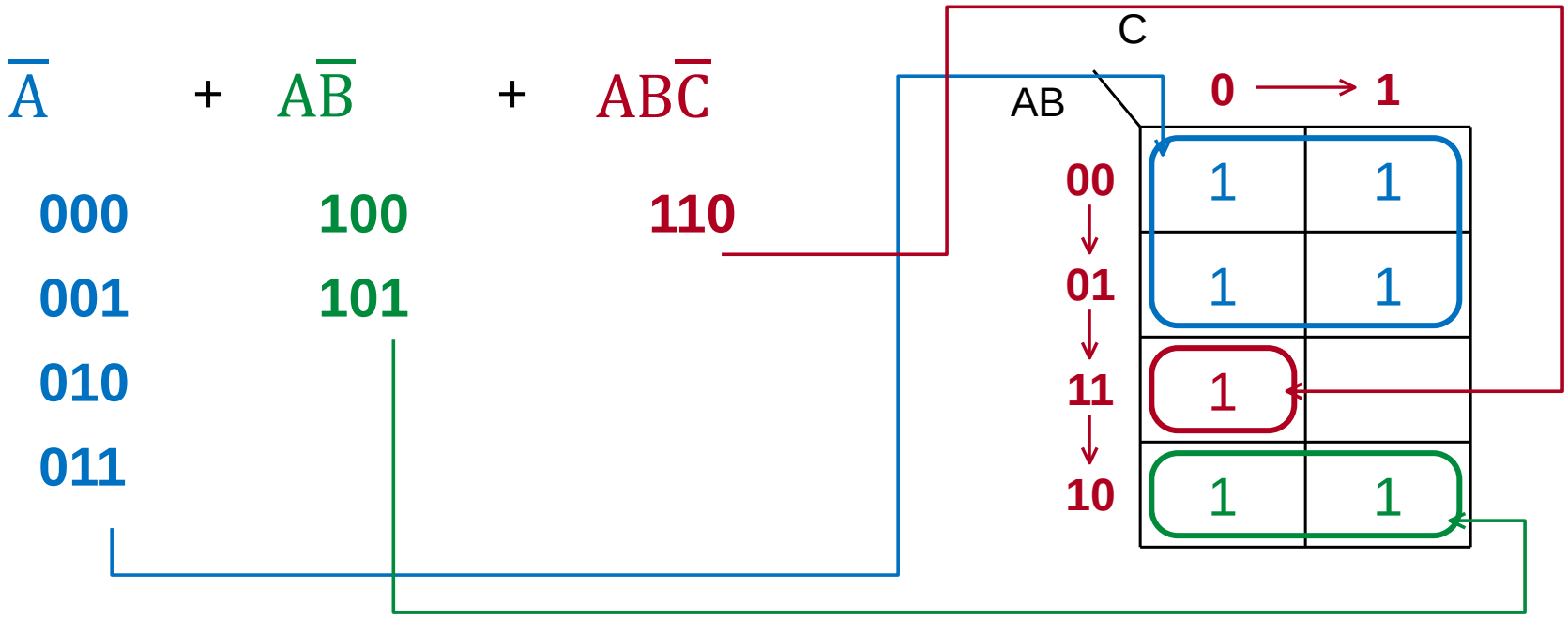
- For a standard SOP, place a 1 on the Karnaugh map in the cell having the same value as the product term.



Karnaugh Map SOP Minimization

Mapping a Non-Standard SOP Expression

- Convert it to standard form by **numerical expansion**.
- For each missing variable, the binary value of the product term is split into two by attaching a 1 and 0 respectively.

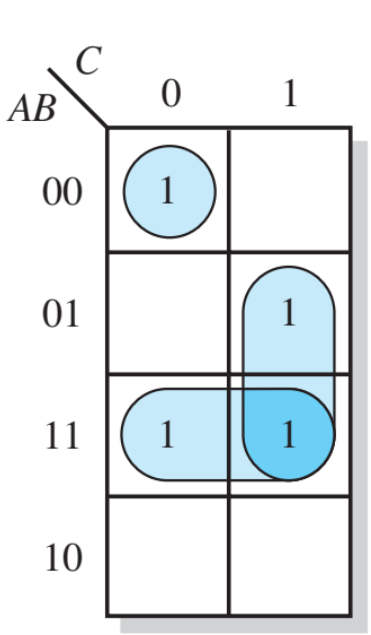


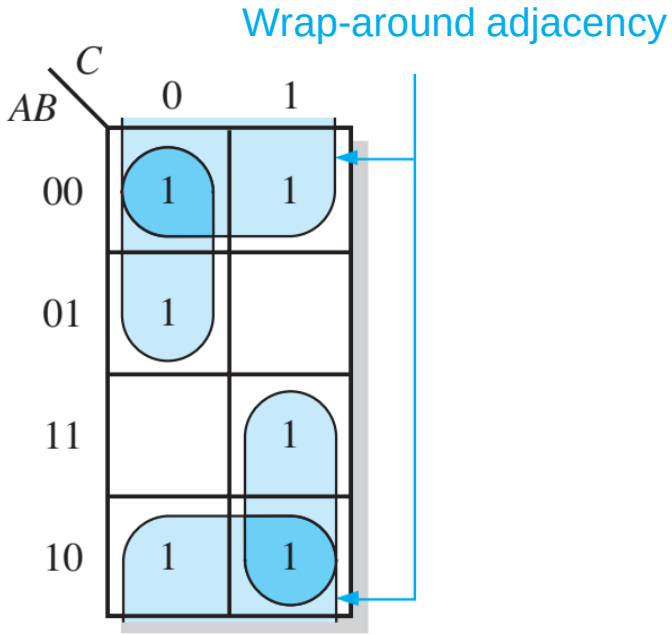
Karnaugh Map SOP Minimization

Step 1: Grouping the 1s

The goal is to **maximize the size** of the groups (shorter product terms) and to **minimize the number** of groups (less product terms).

- A group may contain 1, 2, 4, 8, or 16 (2^n) adjacent cells.
- Each 1 must be included in one or **more** groups.

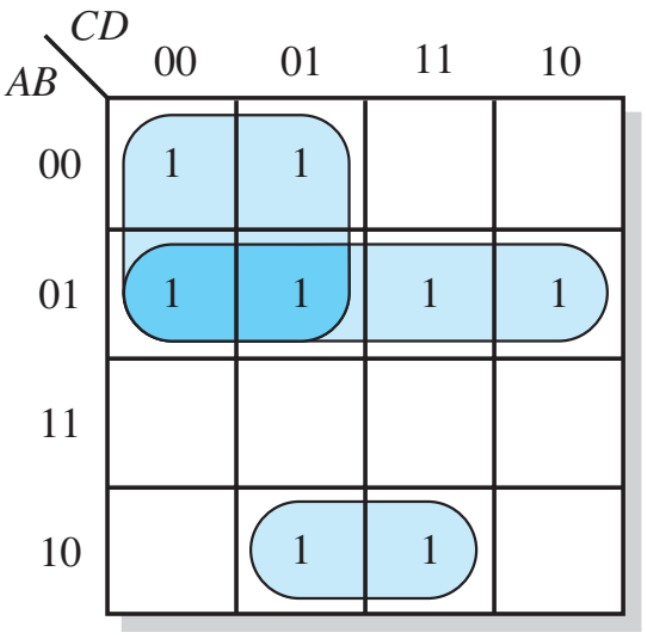


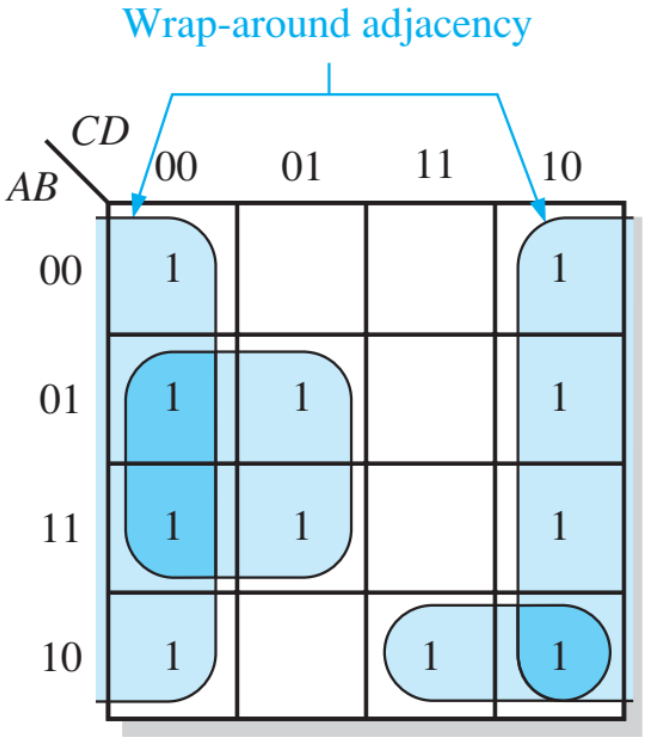


Karnaugh Map SOP Minimization

Step 1: Grouping the 1s

- In some cases, there may be more than one way to group the 1s to form maximum groupings.



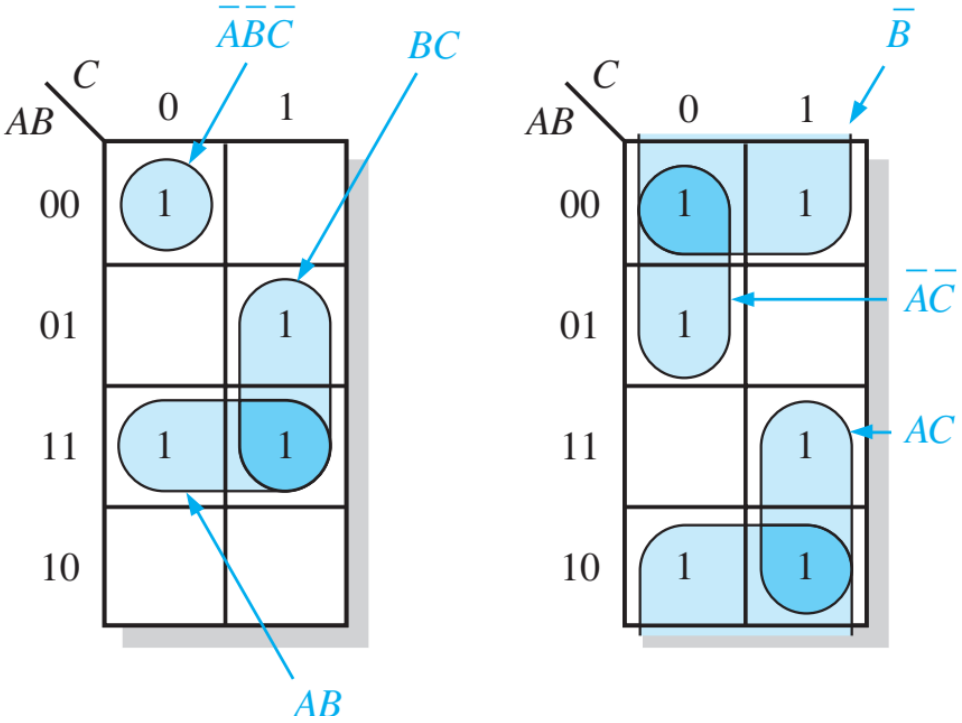


Karnaugh Map SOP Minimization

Step 2: Determine the Minimum SOP

- When a variable appears in both complemented and uncomplemented form in a group, that variable is **eliminated**.

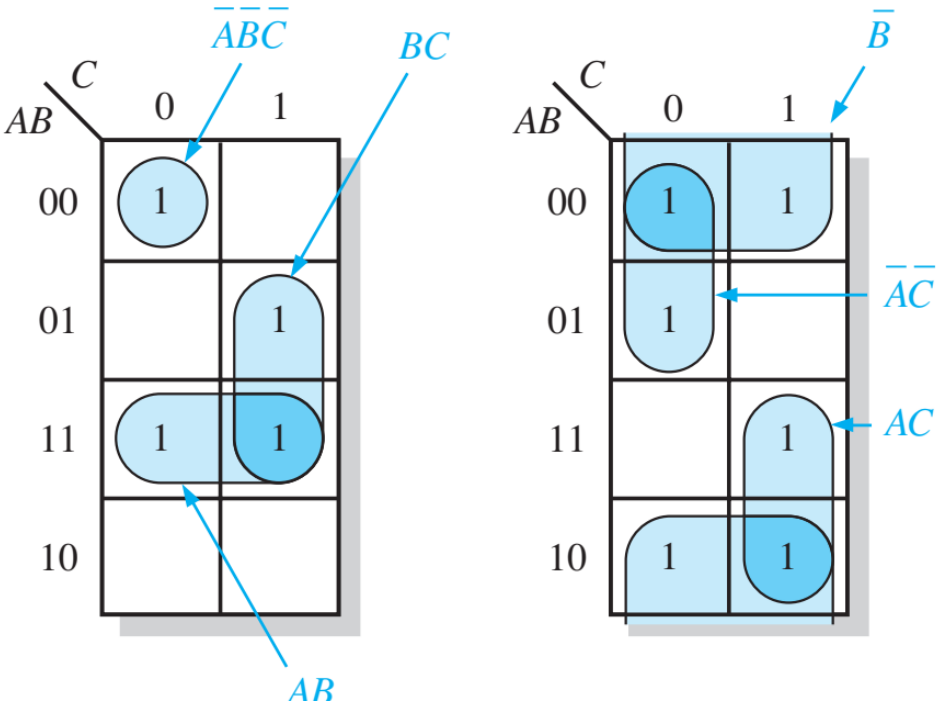
- Variables that are the same for all cells of the group must appear.



Karnaugh Map SOP Minimization

Step 2: Determine the Minimum SOP

- The variable that is **1** for all cells of the group appears in **uncomplemented form**.
- The variable that is **0** for all cells of the group appears in **complemented form**.



Karnaugh Map SOP Minimization

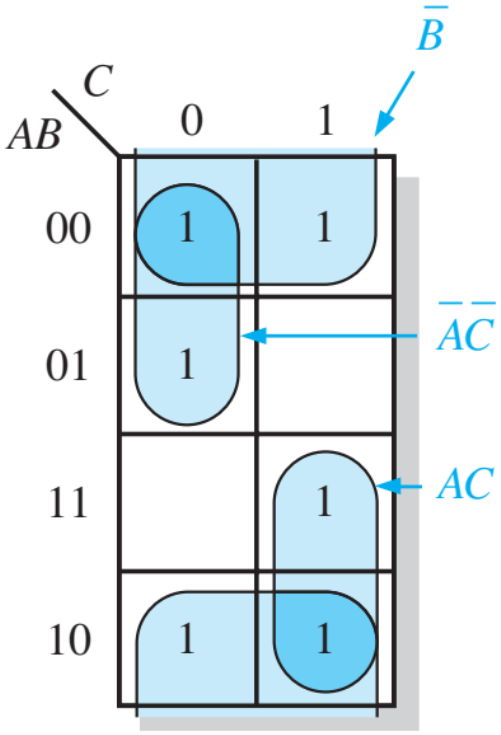
Step 2: Determine the Minimum SOP

For a 3-variable map:

1. A 4-cell group yields a 1-variable term.

2. A 2-cell group yields a 2-variable product term.

3. A 1-cell group yields a 3-variable product term.



The larger a group, the shorter a product term.

Practice 2

For ABC, group 010, 011, 110 and 111. Which variable remains?

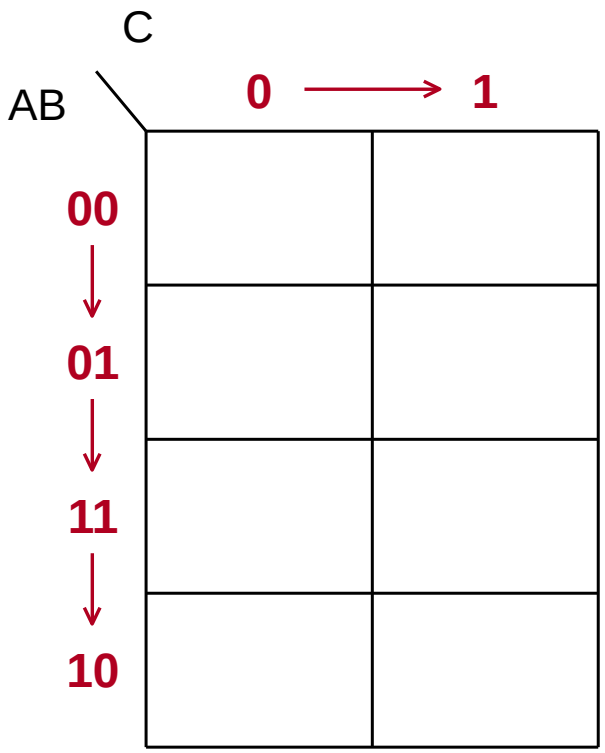
Practice 3: SOP Mapping

Map the SOP expression on a Karnaugh map.

$$BC + \overline{A}\,\overline{C}$$

1. Expand each product term numerically.

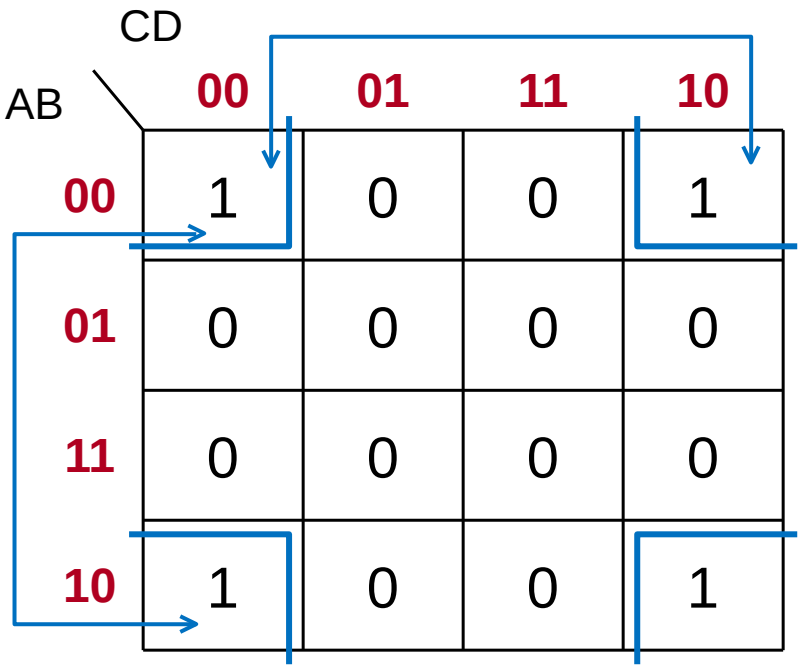
2. Place a 1 in every corresponding cell.



Karnaugh Map SOP Minimization

Four-corner Group

The **top row** is adjacent to the **bottom row**.
The **leftmost column** is adjacent to the **rightmost column**.



One group of four cells

B = 0 and D = 0 in all four cells.
A and C change.

$$F = \overline{B}\,\overline{D}$$

Karnaugh Map SOP Minimization

“Don’t Care”

• Sometimes some input variable combinations will never occur, e.g. six invalid numbers in BCD code.

• Either a 1 or a 0 may be assigned to the output. They can be treated as “don’t care” terms, written as X.

X: output is unspecified for this input combination.

A	B	C	D	Y
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	X
1	0	1	1	X
1	1	0	0	X
1	1	0	1	X
1	1	1	0	X
1	1	1	1	X

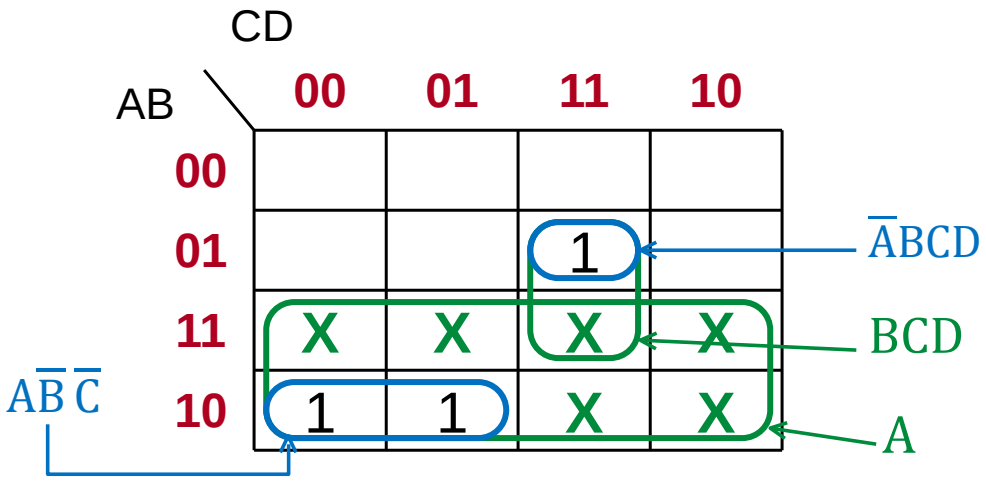
Karnaugh Map SOP Minimization

"Don't Care"

A	B	C	D	Y
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	X
1	0	1	1	X
1	1	0	0	X
1	1	0	1	X
1	1	1	0	X
1	1	1	1	X

Can be used to simplify Boolean expressions.

When grouping the 1s, the Xs can be treated as 1s to **make a larger grouping** or as 0s if they **cannot be used to advantage**.



Without X: $Y = A\overline{B}\,\overline{C} + \overline{A}BCD$

With X: $Y = A + BCD$

Karnaugh Map SOP Minimization

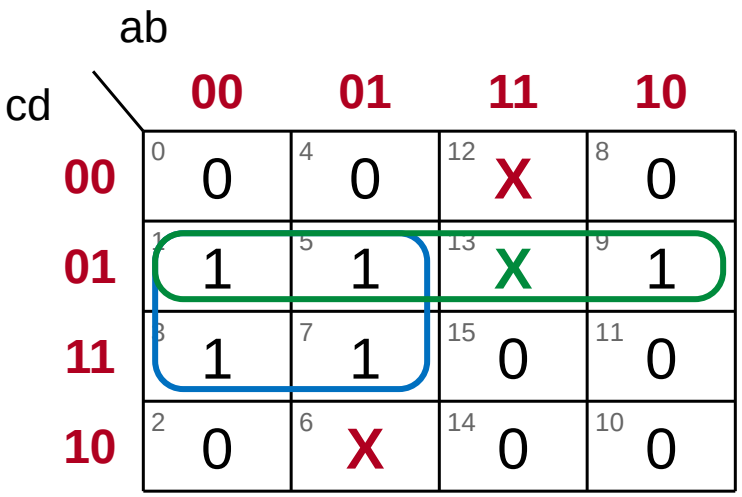
"Don't Care": Selective Use of X

a	b	c	d	Y
0	0	0	0	0
0	0	0	1	1
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	X
0	1	1	1	1
1	0	0	0	0
1	0	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	0	0	X
1	1	0	1	X
1	1	1	0	0
1	1	1	1	0

All the 1s must be covered, but the Xs are **only used if they will simplify** the resulting expression.

Columns: ab. Rows: cd.

1s: 1, 3, 5, 7, 9 Xs: 6, 12, 13



$f = \bar{a}d + \bar{c}d$

Only X13 is used. X6 and X12 are not used.

Practice 4: Don’t-care Terms

A	B	C	D	Y
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	**1**
1	0	0	0	**1**
1	0	0	1	**1**
1	0	1	0	**X**
1	0	1	1	**X**
1	1	0	0	**X**
1	1	0	1	**X**
1	1	1	0	**X**
1	1	1	1	**X**

For the BCD example:

1. Can 1110 be included in a group of 1s? Why?

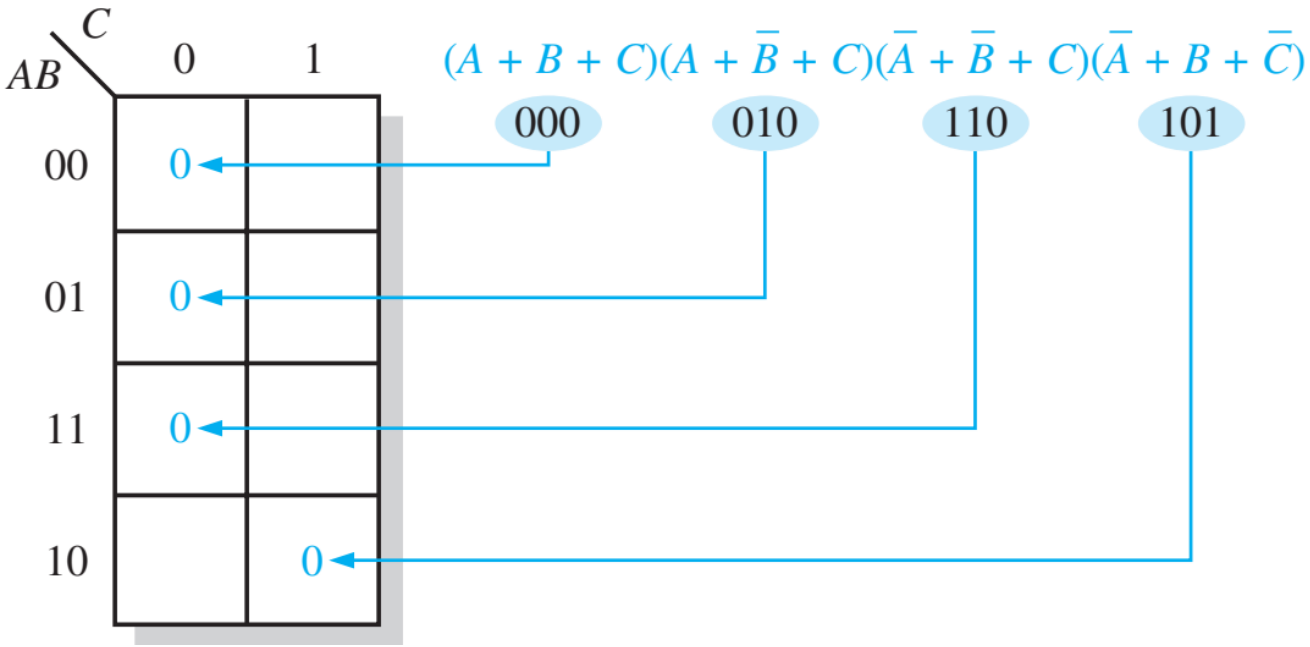
2. Can 0110 be included in a group of 1s? Why?

3. Must every X be covered by a group?

Karnaugh Map POS Minimization

Mapping a Standard POS Expression

- For a standard POS, place a 0 on the Karnaugh map in the cell having the same value as the sum term.



Karnaugh Map POS Minimization

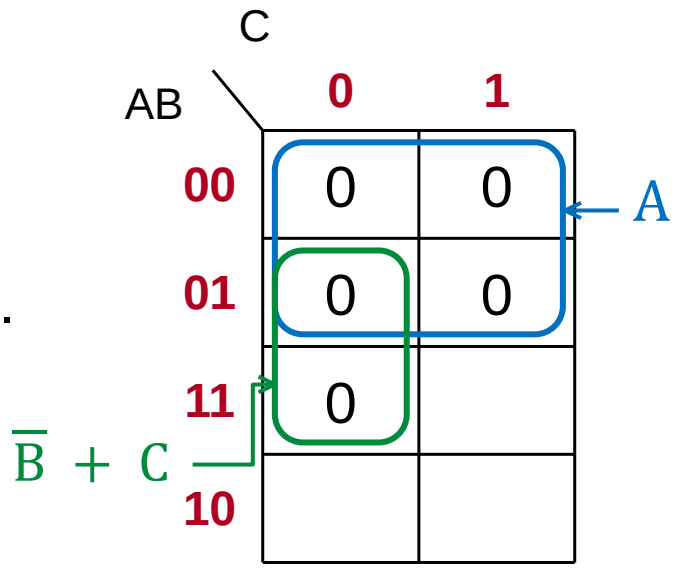
Karnaugh Map Simplification

- Same as for an SOP except grouping 0s.
- The binary values of the sum terms are 000, 001, 010, 011, 110.

$$(A + B + C)(A + B + \overline{C})(A + \overline{B} + C)(A + \overline{B} + \overline{C})(\overline{A} + \overline{B} + C)$$

If a variable is always 0, it appears in uncomplemented form;

if it is always 1, in complemented form.



The minimum POS is

$$A(\overline{B} + C)$$

Practice 5: SOP and POS

For the two marked groups:

1. Write the SOP product term for the blue group of 1s (101 and 111).

2. Write the POS sum term for the green group of 0s (010 and 110).

Variable order: A, B, C

